

Batch ethanol recovery from winterized botanical extract with AUTOaccurate

Increased Efficiency in Solvent Recovery using the Heidolph Hei-VAP Industrial Package

Ethanol is one of the most preferred solvents for extraction and winterization processes. The main reasons are the low toxicity, and the easy handling compared to other methods and solvents. Since it is, mostly due to taxes, one of the more expensive solvents, recycling and re-use of the ethanol is very important to minimize production costs and do have a positive effect on the environmental footprint of companies. Many manufacturers also do not want to deal with finding the perfect vacuum settings and therefore prefer automatisms that guarantee a high-performance distillation process for solvent recovery. This whitepaper shows how the ethanol of a batch of winterized botanical extract was effectively recovered with Heidolph's automatic boiling point detection AUTOaccurate.

Keywords: Solvent recovery, ethanol, automatic, distillation, Hei-VAP, industrial rotary, AUTOaccurate.

Introduction

Ethanol that has been recovered from a winterized batch of botanical extract can be re-used for the extraction and winterization processes for several times. Since batch evaporation is a dynamic process, a performant system requires a frequent adjustment of the vacuum values during the separation, to ensure a highly productive workflow and a high-quality result with no residual ethanol present in the botanical extract. For unexperienced users it can be a challenge to find and to maintain the right parameter settings of their evaporator that on the one hand guarantee for a good separation and on the other hand work in an efficient way.

New options due to automated process control

Many manufacturers try to meet this challenge by hiring one or more person just to have an eye on the evaporator

and to adjust the vacuum values when necessary. Those technicians often are not experienced in working with evaporators and run the machine with low efficiency, decreasing the daily output remarkably.

With the Hei-VAP Industrial packages, Heidolph offers a system solution that can increase efficiency, costs and product quality. With the AUTOaccurate function, the perfect vacuum setting is found, held and adjusted to the changes during the whole evaporation process. It omits the necessity to constantly check, if the process runs performant, because the AUTOaccurate sensor ensures optimal efficiency of the condenser's capacity. After the evaporation of the solvent, the vacuum is further lowered to a preset final pressure, to ensure that all solvent has been recovered. After the process, the evaporator automatically goes into a safe condition by switching off and lowering the heating bath, stopping the rotation and aerating the system.

Method: Successful batch recovery of ethanol from winterized botanical extract using AUTOaccurate

The Heidolph Hei-VAP Industrial package was used to recover the ethanol from a batch of winterized botanical extract using the AUTOaccurate function. Therefore, the recirculating chiller was set to 5 °C, the heating bath was pre-heated to 60 °C and the rotation speed set to 130 rpm. In the AUTOaccurate settings a Delta T of 5 °C was chosen and a final pressure of 30 mbar. To facilitate the filling of the evaporation flask, and omit the risk of spilling, the replenishment valve was used instead of detaching the flask and fill it manually. With a vacuum setting of 200 mbar in the Set Pressure mode a batch size of 8 l was introduced into the evaporation flask via opening the replenishment valve. After changing to the AUTOaccurate mode, the process was started. From there on the AUTOaccurate function takes over the regulation of the vacuum setting for optimal conditions until the final pressure is reached. The result is a clear and colourless ethanol distillate, that is ready to be reused for extraction and winterization and a clear, golden coloured botanical extract with no solvent residue.

Conclusion:

Heidolph meets the challenge

This example shows that Heidolph products optimize and facilitate batch ethanol recovery from winterized botanical extracts for the re-use in extraction and winterization. In comparison to manual evaporation working with AUTOaccurate ensures that the process runs performant and with optimal conditions, saving time and resulting in a high-quality product with no residual solvent.



Hei-VAP Industrial

Large Scale Evaporator